Fixation

Parasites were removed from the intestine of the infected species of fishes. The parasites collected were kept for relaxing, upon being fully relaxed parasite specimens were fixed in the fixatives prescribed for different helminthic group i.e. parasites were washed in 0.6% saline solution and then fixed in 5% formalin solution. After fixation parasites were identified, for this purpose they were stained in Alum carmine, dehydrated in glacial acidic acid, then after cleared in methyl salicylate and finally mounted in canada balsam.

RESULTS

Parasites were present in 88% of the fish specimens examined. In which *Clarias batrachus* and *Channa punctatus* were infected with most varieties of parasites, whereas *Catla catla, Cirrhinus mrigala, Mystus vittatus* were infected with only one species of cestode parasites, *Wallago attu* and *Heteropneustes fossilis* were infected with moderate number of species. The cestode infection were higher in *Clarias batrachus, Wallago attu, Channa punctatus, Heteropneustes fossilis* whereas minimum in *Catla catla, Cirrhinus mrigala, Mystus vittatus*. No cestodes were found in *Labeo rohita* probable reason for this may be, they are herbivores.

One of the species of cestode parasites i.e. *Djombangia penetrans* was common in *Clarias batrachus, Cirrhinus mrigala & Heteropneustes fossilis*.

Many researchers have studied about cestode parasite of freshwater fishes throughout the world.[6-10] In India also many workers have worked on the cestode parasitic of freshwater fishes.[11-15]

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REFERENCES

1. **Schonfeldt H. C. and Hall N. 2013.** Fish, chicken, lean meat and eggs can be eaten daily: a food-based dietary guideline for South Africa. *South African Journal of Clinical Nutrition*. **26:**S66-S76.
2. **Helfman Gene S. 2007.** Fish Conservation: A Guide to Understanding and Restoring Global Aquatic Biodiversity and Fishery Resources. Island Press. p. 11. ISBN 978-1-59726-760-1.
3. **He K., Song Y., Daviglus M. L., Liu K., Van Horn L., Dyer A. R. and Greenland P. 2004.** Accumulated evidence on fish consumption and coronary heart disease mortality: a meta-analysis of cohort studies. *Circulation*. **109(22):**2705-2711.
4. **Begon M., Harper J. L. and Townsend C.R. 1990.** Ecology: Individuals, Populations and Communities, 2nd edition. Blackwell Scientific Publications, Boston, MA, 945 pp.
5. **Amlacher E. 1970.** Textbook of fish diseases, (Engl.Transl.). T.F.H. Publ., Jersey City.
6. **Alam M. J., Rakibuzzaman M., Hasan M. M. 2010.** Comparative study of endo-parasitic infestation in *Channa punctatus* (Bloch, 1793) collected from Hatchery and Sewage lagoon. *Nat Sci.* **8:**152–156.
7. **Baylis B. A. 1933.** On some parasitic worms from Java with remarks on the Acanthocephalan genus *Pallisentis*. *Ann Mag Nat His.* **10:**443–449.
8. **Cohn L. 1908.** Die Anatomic eines neuen Fischcestoden. Centralblatt fur Bakteiologie, Parasitenkunde, Infection-shrank-heiten und Hygiene. *Abteilung L Originale.* **46:**134–139.
9. **Dollfus R. P. 1934.** Sur un cestode Pseudophyllidae parasite de Poisson d' Ornement. *Bull Soc Zool Fr.* **59:**476–490.
10. **Fuhrmann O., Baer J. G. 1925.** Zoological results of the third Tanganyika expedition conducted by Dr. W. A. Cunnington, 1904–1905. Report on the Cestodes. In: *Proceedings of the Zoological Society of London*, pp 79–100
11. **Gairola D, Malhotra SK. 1986.** Cestode fauna of food fishes in river Ganges around an Indian subhumid region. I. *Senga gangesii* n. sp. from *Mystus vittatus*. *Jpn Journal Parasitol.* **36:**471–474.
12. **Gupta V, Parmar S. 1986.** On a new cestodes, *Senga indica* sp. nov. from the intestine of the freshwater fish *Mastacembelus armatus* (Lacep) from Lucknow. *Indian Journal Helminthol.* **37:**96–99.

13. **Hasnain M. 1922.** On a new cestode *Senga chauhani* sp. nov. from a fish host *Channa punctatus* from Jamshedpur. *Indian J Helminthol.* **44:**123–127.

14. **Hiware C. J. 2010.** The occurrence of cestode parasites collected from some freshwater fishes of Maharashtra state, India. *Trend Res Sci Technol.* **2:**31–38.

15. **Jadhav B. V., Ghavane A. B., Jadhav A. P. 1991.** Two new Pseudophyllidean cestodes from *Mastacembelus armatus* at Daryapur (M.S) India. *Riv Parasitol.* **8:**19–22.

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Synthesis, spectral characterization and antimicrobial activity of Chromium (III) complex with 2-Amino phenol

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Abstract- A series of 2-aminophenols including methoxy group were synthesized using appropriate synthetic route. The structures of the compound were determined with spectroscopic techniques and analytical methods. IR spectra of the compounds including and not including hydrogen bonding were compared. The compounds 2 and 4 show the characteristic UV bands attributed to the NH-forms. On the other hand, antibacterial and antifungal activities of the compounds were investigated. Most of the compounds show selective activity toward *Staphylococcus epidermidis* and *Candida albicans*.

Key words: *Staphylococcus epidermidis, Candida albicans*, Chromium, UV bands

INTRODUCTION

2-Amino phenol is an organic compound having chemical formula C_6H_7NO with molar mass of 109.13g/mole. Along with its isomer 4-Amino phenol, it is an amphoteric molecule and a reducing agent.[1] It is useful reagent for the synthesis of dyes and hetrocyclic compound. Reflecting its slight hydrophilic character, white powder is moderately soluble in alcohol and can be recrystallized from hot water. 2-Amino phenol becomes quite popular in coordination chemistry because of its Schiff bases, which generate a rich variety of complexes.[2] Recently, the oxidation of many organic compounds with ditertiary butyl chromate has become a subject of great interest and many research works has been published in this present paper, we have reported the synthesis and study of some Chromium (III) complexes of 2-Amino phenol by oxidizing it with TBC to explore the versatility of the oxidizing agent.

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MATERIALS & METHODS

Preparation of reaction mixture

Substrate/oxidant ratio 1:1-1gm CrO_3 dissolved in 10 ml of tert butyl alcohol to prepare tert butyl chromate and 1gm 2-Amino phenol dissolved in 10 ml THF.

Substrate/oxidant ratio 1:2-1gm CrO_3 dissolved in 10 ml of tert butyl alcohol to prepare tert butyl chromate and 2.4gm 2-Amino phenol dissolved in 10 ml THF.

Substrate/oxidant ratio 1:3-1gm CrO_3 dissolved in 10 ml of tert butyl alcohol to prepare tert butyl chromate and 3.16gm 2-Amino phenol dissolved in 10 ml THF.

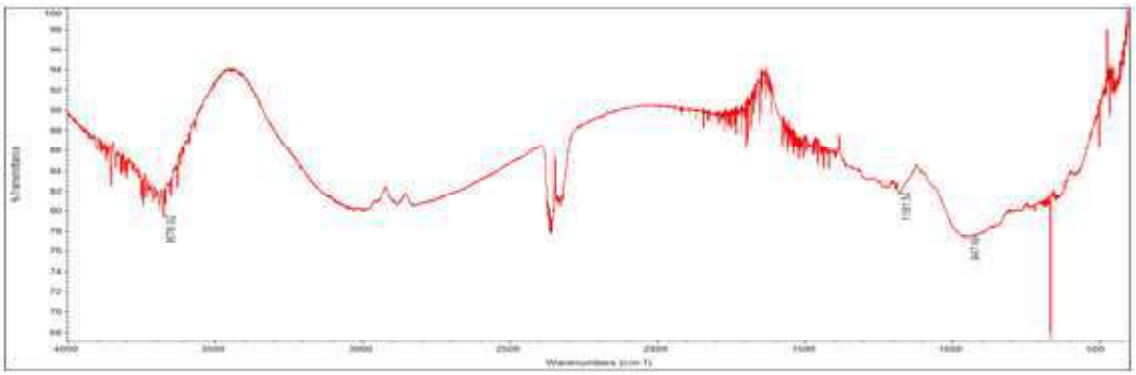
Procedure

2-Amino phenol was found to be freely soluble in selected solvents namely THF and 1,4-dioxane. Oxidation of this substrate with TBC was performed in two sets, each for the two solvents.[3] A solution of the oxidant TBC was prepared separately in situ from Chromium trioxide and tertiary butanol, varying the substrate/oxidant molar ratio as 1:1, 1:2 and 1:3 for each set. The two solution were mixed together to prepare a homogeneous reaction

mixture, stirred continuously for a considerable time using a magnetic stirrer and any change in consistency was noted. The mixture was heated in microwave oven.[4] Reaction were found to be exothermic, in general, except for those in 1,4-dioxane, in which they were endothermic as indicated by a drop in temperature of the reaction mixture. The six products formed in each of these cases were washed with acetone, dried and weighed.

Spectral character

Absence of characteristic C-H stretching frequency for an aldehyde group suggests that the amino phenol aldehyde group has been oxidized to a carboxylic group. This is further confirmed by the appearance of relevant peaks near 761.02 cm^{-1} for (C-O) stretching vibrations of a carboxylic group. The characteristic peaks for aromatic (C-H) stretching, aromatic (C-C) and (C=C) in-ring stretching. The presence of Cr-O and Cr=O bonds in these complexes is indicated by the (Cr-0) stretching frequencies ranging from 478.65-454.10 cm^{-1} and (Cr=0) peaks near 761.02, 667.55 cm^{1}. The broad and strong absorption bands appearing from 1161.69 947.69 cm^{1} shows the presence of the H-Bonded (O-H) group, intensity and peak broadening varying greatly with variation in substrate/ oxidant ratio. This indicates the difference in the extent of coordination of the (O-H) oxygen atom with the Cr atom.[5] The lowest (O-H) stretching frequency is centred at 3341.10 cm^{-1} in the compound labelled AP-101, while the maximum value recorded is for AP-111 at 3676 cm^{-1}. A strong (C-O) carbonyl absorption is visible near 1369.46 cm^{-1}. However, the vibrations near 1574.87-1556.97 cm^{-1}, 1488.19-1393 cm^{-1} and 1283.43-1203.11 cm^{-1}, respectively, indicate that the aromatic ring in amino phenol is intact and has not undergone any decomposition/ cleavage during the course of oxidation. The presence of (C-O) stretching vibrations for the methoxy group near 667.56 cm^{-1} in the IR spectrum of these complexes is indicative of the oxygen atom of the other group of Amino phenol being uncoordinated. This is supported by the weak alkyl (C-H) stretching vibrations appearing near 1147.07 cm^{-1} for each of these compounds. Other weak absorption bands for bending and rocking vibrations of (C-H) group were recorded near 686 cm^{1} and 478.65-454.10 $cm^{-1,}$ respectively.[6] A comparative spectral analysis for the six oxidation products of Aminophenol shows a considerable shift in the relative intensities and peak positions of these vibrational frequencies, which may be attributed to a difference in the percentage composition of Cr metal and variable extent of its coordination with the ligand. The strong peaks at 3341.22 cm^{1} and 2373 cm^{-1} are probably due to the impurities present in the KBr. A very weak peak near 3676.92 cm^{1} is characteristic of CO, absorption.



Antimicrobial Activity

This section will describe the antibacterial activities of Cr (III) complexes as well as their ligands.[1-3,7] It has also been proposed that chelation decreased the polarity of the metal ion mainly due to partial sharing of its positive charge with a donor group throughout the entire chelate ring system. This chelation process, therefore, enhanced the lipophilic nature of the central metal ion, which in turn facilitated its permeation through the lipid layer of the membrane, thereby allowing the metal complex to more easily cross the bacterial membrane, increasing the complexes' activity. In addition, many other factors, such as solubility, dipole moment, conductivity that influenced by a metal ion, could contribute to the antibacterial activity of the complex. On chelation, the delocalisation of pi-electrons over the whole chelate ring increases the liposolubility of the complexes. The hydrocarbon tail